UNIT II- SIZE REDUCTION

Size reduction is applied to all ways in which particles of solids are cut or broken into smaller pieces. The objective of crushing and grinding is to produce small particles from larger ones. Smaller particles are desired because of their larger surface. In most reactions involving solid particles the rate is directly proportional to the area of contact with the second phase. In leaching, the rate of leaching is increased if the area of contact is more between solvent and the solid to be extracted. In addition the distance the solvent has to penetrate into the particles in order to gain access to more remote pockets of solute is also reduced. This factor is also important in drying porous solids, where reduction in size causes both an increase in area and a reduction in the distance, the moisture must travel within the particle in order to reach the surface. More intimate mixing of solids can be achieved if the particle size is small.

Solids may be broken by applying different forces. Four forces commonly used in size reduction machines are (1) Compression (2) Impact (3) Attrition or rubbing (4) Cutting.

Ex: Nut cracker, hammer, file, pair of shears.

Ex: Type of Force	Every Day Example	Typical Machine (Tools)
Compression	Cracking of net with a nut cracker.	Jaw or Gyratory crusher.
Impact	Striking of stone with hammer , pounding.	Hammer mill and Impactor.
Attrition	Sharpening of knife on an emery wheel.	Rod mills and ball mills.
Shredding	Plaining of wood with Jack plane.	Cutters and shredders.

Compression is used for coarse reduction of hard solids to give relatively few fines. Impact gives coarse, medium or fine particles. Attrition yields very fine products from soft, non-abrasive materials. Cutting gives a definite particle size and sometimes a definite shape with few or no fines.

Requirements of an Ideal Crusher and Grinder:

1. It should have a large capacity.

2. Should consume less power per unit of product.

3. Should yield a product of single size or size distribution desired.

Unlike an ideal crusher or grinder an actual crusher or grinder does not yield a single size product. The product always consists of a mixture of particles. Some machines (especially in the grinder class) are designed to control the size of the largest particle in their products, but the fine sizes are not under control. In some types of grinders fines are minimized but they are not eliminated.

Power requirement in Comminution:

The cost of power is a major expense in crushing and Grinding, so the factors that control this cost is important. During size reduction new surface is created.

Crushing Efficiency: 'η_c'

The ratio of the surface energy created by crushing to the energy absorbed by the solid is the crushing efficiency 'η_c'.

$$\eta_C = \frac{Surface\ energy\ created\ by\ crushing}{energy\ absorbed\ by\ the\ solid}$$

$$\eta_C = \frac{e_s(A_{Wb} - A_{Wa})}{W_n}$$

$\eta_C = crushing\ efficiency$
$e_s = surface\ energy\ per\ unit\ Area, {}^{J}/_{m^2}$
$A_{wb} = area\ of\ product\ m^2$
$A_{wa} = area\ of\ feed\ m^2$
$w_n = energy\ absorbed\ by\ material, {}^{J}/_{kg}$
$\dot{m} = mass\ feed\ rate\ {}^{kg}/_{s}$
$\rho_P = Density\ of\ particle$

The surface energy created by fracture is small in comparison with the total mechanical energy stored in the material at the time of rupture. Most of the mechanical energy stored is converted into heat. Crushing efficiencies are therefore low.

Mechanical Efficiency: 'η_m':

The energy absorbed by the solid w_n is less than that fed to the machine, part of the total energy input w is used to overcome friction in the bearing or other moving parts and the rest is available for crushing.

The ratio of the energy absorbed to the energy input is the mechanical efficiency η_m.

$$\eta_m = \frac{Energy\ absorbed}{Energy\ Input} = \frac{w_n}{w}$$

$$\eta_m = \frac{w_n}{w} = \frac{e_s(A_{wb} - A_{wa})}{\eta_c w} \Rightarrow \quad w = \frac{w_n}{\eta_c} = \frac{e_s(A_{wb} - A_{wa})}{\eta_m \eta_c}$$

If $\dot{m}$ is the feed rate, the power required by the machine

$$P = w\dot{m} = \dot{m}\ \frac{e_s(A_{wb} - A_{wa})}{\eta_m \eta_c}$$

We know that

$$A_{wb} = specific\ area\ of\ product\ m^2 = \frac{6}{\phi_S \rho_P \overline{D_{Sb}}}$$

$$A_{wa} = specific\ area\ of\ feed\ m^2 \frac{6}{\phi_S \rho_P \overline{D_{Sa}}}$$

$$P = \frac{6\dot{m}e_s}{\rho_P \eta_m \eta_c}\left[\frac{1}{\phi_b \overline{D_{Sb}}} - \frac{1}{\overline{\phi_a D_{Sa}}}\right]$$

Where $\bar{D}_{sb}$ & $\bar{D}_{sa}$ are the volume surface mean dia. of the product and feed respectively.

Laws of Comminution:

Rittinger's Law: (1867):

The law states that the work required in crushing is proportional to the new surface created.

$$P = w\dot{m} = \dot{m}\,\frac{e_s(A_{wb} - A_{wa})}{\eta_m \eta_c}$$

$$\frac{P}{\dot{m}} = K_R[A_{wb} - A_{wa}]$$

$$A_{wb} = specific\ area\ of\ product\ m^2 = \frac{6}{\phi_b \rho_P \overline{D_{Sb}}}$$

$$A_{wa} = specific\ area\ of\ feed\ m^2 \frac{6}{\phi_a \rho_P \overline{D_{Sa}}}$$

$$P = \frac{6\dot{m}e_s}{\rho_P \eta_m \eta_c}\left[\frac{1}{\phi_b \overline{D_{Sb}}} - \frac{1}{\overline{\phi_a D_{Sa}}}\right]$$

$$\frac{P}{\dot{m}} = \frac{6e_s}{\rho_P \eta_m \eta_c}\left[\frac{1}{\phi_b \overline{D_{Sb}}} - \frac{1}{\overline{\phi_a D_{Sa}}}\right]$$

If $\phi_{a\,a}$ and ϕ_b are equal and the mechanical efficiency is constant, crushing efficiency is constant. The various constants can be combined into a single constant K_R .

$$\frac{P}{\dot{m}} = K_R\left[\frac{1}{\overline{D_{Sb}}} - \frac{1}{\overline{D_{Sa}}}\right],$$

$$K_R = Rittingers\ Constant = \frac{6e_s}{\rho_P \eta_m \eta_c},$$

$$Rittingers\ constant = \frac{1}{Rittingers\ number}$$

$$\frac{P}{\dot{m}} = K_R[A_{wb} - A_{wa}] = \frac{[A_{wb} - A_{wa}]}{Rittingers\ number}$$

Or

$$Rittingers\ number = \frac{[A_{wb} - A_{wa}]}{\frac{P}{\dot{m}}}$$

Note: Rittingers number speaks about new surface created to energy consumed. Higher the Rittingers number, lesser is the power consumed and better grindability.

Limitations: Mechanical losses due to friction and inertia in grinding equipment is not accounted in Rittingers law. It speaks about only the new surface area created. Some energy is also required to run the equipment empty which is called mechanical losses. Mechanical losses is not accounted.

$$Rittingers\ Constant = \frac{6e_s}{\rho_P \eta_m \eta_c}$$

$$Overall\ Energy\ Efficiency\ = \frac{Energy\ required\ to\ create\ new\ surface}{Total\ energy\ supplied}$$

$$Over\ all\ or\ mechanical\ efficiency\ \eta_m = \frac{w_n}{w},$$

$$Theoretical\ Efficiency = \frac{Energy\ required\ to\ creatge\ new\ surface}{Total\ energy\ supplied - energy\ required\ to\ run\ an\ empty\ mill}$$

Rittingers law is applicable mainly to that part of the process, where new surface is being created and holds most accurately for fine grinding where the increase in surface per unit mass of material is predominant. Thus, it gives better results for fine grinding where there is much greater change in surface area. **This law is applicable for feed size of less than 0.05 mm**

Kick's Law: (1885):

Kick's law is based on stress analysis of plastic deformation within the elastic limit, which states that the work required for crushing a given mass of material is constant for the same reduction ratio, that is the ratio of the initial particle size $\overline{D_{Sa}}$ to the final particle size $\overline{D_{Sb}}$. This leads to the relation.

$$\frac{P}{\dot{m}} = K_k ln \frac{\overline{D_{Sa}}}{\overline{D_{Sb}}}, \qquad K_k = Kicks\ Constant$$

Both the laws have been shown to apply over limited ranges of particle size, provided K_K and K_R are determined experimentally by tests in a machine of the type to be used and with the material to be crushed. They thus have limited utility and are mainly of historical interest.

Note: The energy required for reducing a 200 mm particle to 50 mm size will be the same as that required for reducing a 2 mm particle to 0.5 mm. But higher amount of energy is required for reducing fine particles to still finer size than for breaking down large pieces of rock

Limitations: Kicks law can be applied for coarse crushing where the feed size is quite large and the reduction ratio is quite low

Note: Kicks Law and Rittingers law have been shown to apply over limited ranges of particles size, provided K_K and K_R are determined experimentally by tests in a machine of the type to be used and with the material to be crushed. They thus have limited utility.
Kicks Law: Coarse feed
Rittingers Law: Fine grinding
Bonds Law intermediate grinding

Bond crushing Law and Work Index: (1952):

A somewhat more realistic method of estimating the power required for crushing and grinding.

The law states that the work required to form particles of size D_P from a very large feed is proportional to the square root of the surface to volume ratio of the product, $\frac{S_P}{V_P}$

$$\frac{P}{\dot{m}} \alpha \sqrt{\frac{S_P}{V_P}}$$

W.k.t $\phi_S = \frac{6v_P}{D_P S_P}$ or $\frac{S_P}{V_P} = \frac{6}{\phi_s D_P}$

$$\therefore \frac{P}{\dot{m}} \alpha \sqrt{\frac{6}{\phi_s D_P}} \quad or \quad \frac{P}{\dot{m}} = \frac{K_B}{\sqrt{D_P}}$$

Where K_B is a constant which depends on the type of machine and on the material being crushed.

$$\therefore \frac{P}{\dot{m}} = K_B \left[\frac{1}{\sqrt{D_{Sb}}} - \frac{1}{\sqrt{D_{Sa}}}\right]$$

To use above equation, a work index W_i is defined. It is defined as the gross energy requirement in kWh / ton of feed needed to reduce a very large feed to such a size that 80 % of the product passes a 100 μm screen.

This definition leads to a relation between K_B and W_i. As it is mentioned as very large feed $\frac{1}{\sqrt{D_{Sa}}}$ *will be very less value and can be neglected*
If D_P in mm, $100\mu m = 100\ x\ 10^{-6} m = 100\ x\ 10^{-6} x 10^{3} mm = 0.1\ mm$
P in kW, and $\dot{m}$- tons / hour.

$$\therefore \frac{P}{\dot{m}} = K_B \left[\frac{1}{\sqrt{D_{Sb}}}\right] = \frac{K_B}{\sqrt{0.1}}$$

$$W_I = \frac{K_B}{\sqrt{0.1}} = \frac{K_B}{0.3162}$$

$$\therefore K_B = = 0.3162\, W_i$$

If 80 % of the feed passes a mesh size of D_{Pa} mm, the above equation can be written as

$$\frac{P}{\dot{m}} = 0.3162\, W_i \left[\frac{1}{\sqrt{\overline{D_{Sb}}}} - \frac{1}{\sqrt{\overline{D_{Sa}}}}\right]$$

The work index is defined as the gross power and it includes the friction (Mechanical energy) in the crusher and the power given by above equation is gross power. Bonds law can be applied for feed size between 0.05 to 50 mm

Note: To use the above formula $\overline{D_{Sb}}$ to be in mm only

Work Index for different Materials

Material	Work Index (kWh/tonne)
Lime Stone	12.74
Gypsum	6.73
Quartz	13.57
Coal	13.00
Bauxite	8.78

These values do not vary much with the equipment and can be applied to wet grinding only. If these values are to be applied for dry grinding, these values are to be multiplied by 4/3.

Generalized relation: All the laws can be derived from a general differential equation.

$$\frac{P}{\dot{m}} = -K\, \frac{d\,\overline{D}}{\overline{D_S}^n}$$

$n = 1$ *gives Kicks Law* & $n = 2$ *gives Rittingers Law* &
$n = 1.5$ *gives Bods Law*

Energy Utilization:

The energy supplied for crushing is utilized in the following ways

1. In producing elastic deformation of the particles before fracture occurs

2. In producing elastic deformation which results in size reduction.

3. In causing elastic distortion of the equipment.

4. In overcoming the friction between particles and between particles and machines.

5. In noise, heat and vibration in the plant.

6. In friction losses in the plant itself.

Relative Applicability of Law of Crushing:

Rittinger's law is applicable mainly to that part of the process where new surface is being created and holds most accurately for fine grinding where the increase in surface per unit mass of material is large.

Kicks law more closely relates to the energy required to effect elastic deformation before fracture occurs, and is more accurate than Rittinger's law for coarse crushing where the amount of surface produced is considerably less.

Bond's law is intermediate between Rittinger's law and Kick's law.

Problems

Pb1 In crushing of iron ore 80% of feed passes through 5mm screen and 80 % product passes through 0.5 mm screen. Now it is desired to alter the product size such that 80 % of the same passes through 0.2 mm screen. What is the % increase in the power consumption.

Solution

$$\left(\frac{P}{\dot{m}}\right)_1 = k_b \left[\frac{1}{\sqrt{\overline{D_{Sb1}}}} - \frac{1}{\sqrt{\overline{D_{Sa1}}}}\right]$$

$$\left(\frac{P}{\dot{m}}\right)_2 = k_b \left[\frac{1}{\sqrt{\overline{D_{Sb2}}}} - \frac{1}{\sqrt{\overline{D_{Sa2}}}}\right]$$

$$\overline{D}_{Sa1} = \overline{D}_{Sa2} = 5\ x\ 10^{-3}\ m$$

$$\overline{D}_{Sb1} = 0.5\ x\ 10^{-3}\ m$$

$$\overline{D}_{Sb2} = 0.2\ x\ 10^{-3}\ m$$

$$\frac{\left(\frac{P}{\dot{m}}\right)_2}{\left(\frac{P}{\dot{m}}\right)_1} = \frac{\left[\frac{1}{\sqrt{\overline{D_{Sb2}}}} - \frac{1}{\sqrt{\overline{D_{Sa2}}}}\right]}{\left[\frac{1}{\sqrt{\overline{D_{Sb1}}}} - \frac{1}{\sqrt{\overline{D_{Sa1}}}}\right]} = \frac{\left[\frac{1}{\sqrt{0.2\ x\ 10^{-3}}} - \frac{1}{\sqrt{5\ x\ 10^{-3}}}\right]}{\left[\frac{1}{\sqrt{0.5\ x\ 10^{-3}}} - \frac{1}{\sqrt{5\ x\ 10^{-3}}}\right]} = \frac{70.71 - 14.14}{44.72 - 14.14}$$

$$= \frac{56.57}{30.58} = \frac{1.85}{1}$$

$$\%\ Increase\ in\ power\ consumption = 85$$

Pb 2. Granite of 1.5 cm size is to be crushed in a crusher. The Rittingers law constant for crushing operation is 10 Jm/Kg. Calculate the energy required for crushing 1 tonn of material.

Mesh No	Average Size Microns	Wt Retained
4/6	3215	4500
6/9	1197	200
9/20	995	10
20/10	326	70
40/60	206	80
60/90	185	20
Pan	-	40

Solution:

$$\boldsymbol{P} = Energy\ required = ? \qquad \frac{P}{\dot{m}} = K_R \left[\frac{1}{\overline{D}_{Sb}} - \frac{1}{\overline{D}_{Sa}}\right]$$

$$m = 1\frac{ton}{s} = 100\ kg/s$$

$$Rittingers\ Constant = 10\ jm/kg$$

$$\overline{D}_{SA} = 1.5\ cm = 1.5\ x\ 10^{-2}\ m$$

$\overline{D}_{SB}\ can\ be\ found\ from\ the\ given\ data\ by\ differential\ analysis,$

$$\overline{D}_{SB} = \frac{1}{\sum\left(\frac{x_i}{\overline{D}_{Pi}}\right)}$$

Mesh No	Average Size Microns $\overline{D}_{Pi}$	Wt Retained	Mass fraction x_i	$\frac{x_i}{\overline{D}_{Pi}}$
6	3215	450	0.517241	0.000160884
9	1197	200	0.229885	0.000192051
20	995	10	0.011494	1.1552E-05
40	326	70	0.08046	0.000246809
60	206	80	0.091954	0.000446379
90	185	20	0.022989	0.000124262
Pan	-	40	0.045977	
		870		$\sum\left(\frac{x_i}{\overline{D}_{Pi}}\right)$ =0.001181937

$$\overline{D}_{SB} = \frac{1}{\sum\left(\frac{x_i}{\overline{D}_{Pi}}\right)} = \frac{1}{0.001181937} = 846.0689\mu = 846\ x\ 10^{-6}m$$

$$\frac{P}{\dot{m}} = K_R \left[\frac{1}{\overline{D_{Sb}}} - \frac{1}{\overline{D_{Sa}}}\right]$$

$$\frac{P}{100} = 10 \left[\frac{1}{846\ x\ 10^{-6}} - \frac{1}{846\ x\ 10^{-6}}\right]$$

$$\therefore \boldsymbol{P} = \ - - - - - - - \boldsymbol{kW}$$

Pb3: A certain crusher accepts a feed material having a volume surface mean diameter of 19 mm and gives a product of volume surface mean diameter of 5 mm. The power required to crush 15 tonnes per hour is 7.5 kW. What will be the power consumption if the capacity is reduced to 12 tonnes per hour.

Solution:

P = 7.5 kW

$\dot{m} = 15\ tonnes/hr$

$\overline{D_{Sb}} = 5mm = 0.005m$

$\overline{D_{Sa}} = 19mm = 0.019m$

$$\frac{P}{\dot{m}} = K_R \left[\frac{1}{\overline{D_{Sb}}} - \frac{1}{\overline{D_{Sa}}}\right]$$

$$\frac{7.5}{15} = K_R \left[\frac{1}{0.005} - \frac{1}{0.019}\right]$$

$$K_R = 3.4\ x\ 10^{-3}$$

For 12 tonnes /hr

$$\frac{P}{12} = 3.4\ x\ 10^{-3} \left[\frac{1}{0.005} - \frac{1}{0.019}\right]$$

$$P = 6\ kW$$

Pb4: What will be the power required to crush 100 tonnes per hour of lime stone if 80 percent of the feed passes 50mm screen and 80 % of the product pases 3.125 mm screen. Work index for lime stone: 12.74

Solution:

P = 7.5 kW

$\dot{m} = 150\ tonnes/hr$

$\overline{D_{Sb}} = 3.125mm$

$\overline{D_{Sa}} = 50mm$

$W_i = 12.74$

$$\frac{P}{150} = 0.3162\ W_i \left[\frac{1}{\sqrt{\overline{D_{Sb}}}} - \frac{1}{\sqrt{\overline{D_{Sa}}}}\right]$$

$$\frac{P}{\dot{m}} = 0.3162\ x\ 12.74 \left[\frac{1}{\sqrt{3.125}} - \frac{1}{\sqrt{50}}\right]$$

$$\boldsymbol{P = 256.4\ kW}$$

Size Reduction Equipments:
Coarse Crusher:

1. Jaw Crusher (Dodge and Blake)

2. Gyratory Crusher.

Intermediate Crushers:

1. Crushing rolls.

2. Hammer mills.

3. Edge runner.

Fine Crusher:

1. Ball mill.

2. Tube mill.

3. Ring roller mill.

Ultrafine Grinders:

1. Fluid energy mill.

2. Agitated mills.

Cutting Machines:

1. Knife Cutters.

Forces and criteria for communition, Laws of size reduction, Work Index. Methods of operating crushers – Free crushing, Choke feeding, Open circuit grinding, Closed circuit grinding, Wet and dry grinding, Equipment for size reduction – Blake jaw crusher, Gyratory crusher, Smooth roll crusher, angle of nip, Impactor, Attrition mill, Ball mill- Critical speed of ball mill

Crushers:

Crushers are slow speed machines for coarse reduction of large quantities of solids. The main types are JAW crushers and Gyratory crushers. These crushers operate by compressing and break large lumps of very hard materials. These are used to break the rocks and ores.

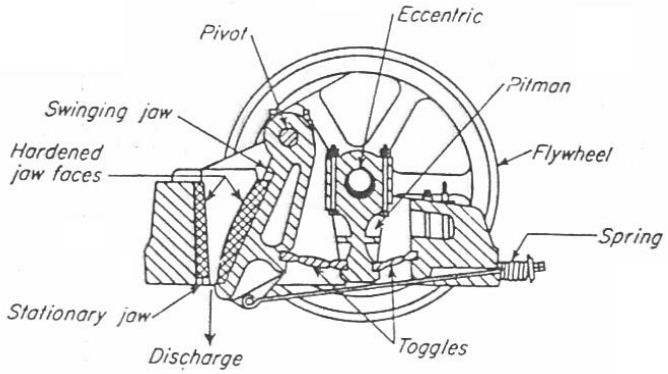
Jaw Crushers:

In a Jaw Crushers feed is admitted between the jaw set to form a 'V' open at the top. One jaw is fixed it is nearly vertical and does not none. This jaw is called axial jaw. The other jaw is the swinging Jaw and this jaw reciprocates in a horizontal plane. The

swinging jaw makes an angle of 20 to 30° with anvil jaw. It is driven by an eccentric so that it applies great compressive force to lumps caught between the jaws. The jaw faces are flat or slightly bulged. They may carry shallow horizontal grooves. Large lumps caught between the upper parts of the jaws are broken, dropped into the narrow space below, and are recrushed the next time the jaw closes. After sufficient reduction they drop out the bottom of the machine. The jaws open and close 250 to 400 times/ min. They are 2 types of Jaw Crushers.

1. Blake Jaw Crushers:

The most common type of jaw crusher is Blake jaw crusher. In this machine an eccentric drives of a pitman connected to 2 toggles, one of which is pinned to the frame and the other to the swinging jaw. The pivot point is at the top of the moveable jaw. The greatest amount of motion is at the bottom of the V, which means that there is a little tendency for a crusher of this kind to choke.



The crushing faces are formed either of manganese steel or of chilled cast iron. The risk of breakage is reduced by grinding the back surface flat or packing with lead. The machine is usually protected so that it is not damaged if lumps of metal in advertently enter, by making one of the toggle plates is the driving mechanism relatively weak so that if any large stresses are setup, this is the first part to fail. Easy renewal of the damaged part is then possible.

Gyratory Crusher:

This type may be looked upon as a Jaw. Jaw crusher with circular Jaws, between materials is being crushed at all times. A conical crushing head Gyrates (none in circle) inside a funnel shaped casing, open at the top. The crushing head is carried on a heavy shaft pivoted at the top of the machine. An eccentric drives the bottom end of the shaft . At any point on the periphery of the casing, therefore the bottom of the crushing head moves towards and then away from the stationary walls. Solids caught in the 'V' shaped space between the head and casing are broken and re broken until they pass out the bottom. The crushing head is free to rotate on the shaft and turns slowly because of friction with the material being crushed.

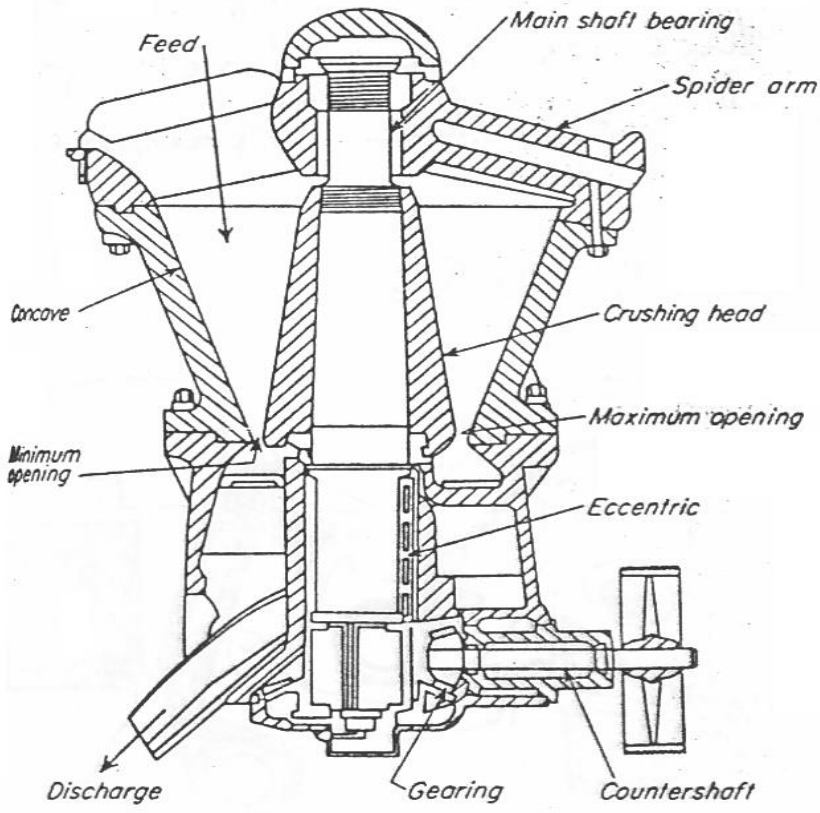
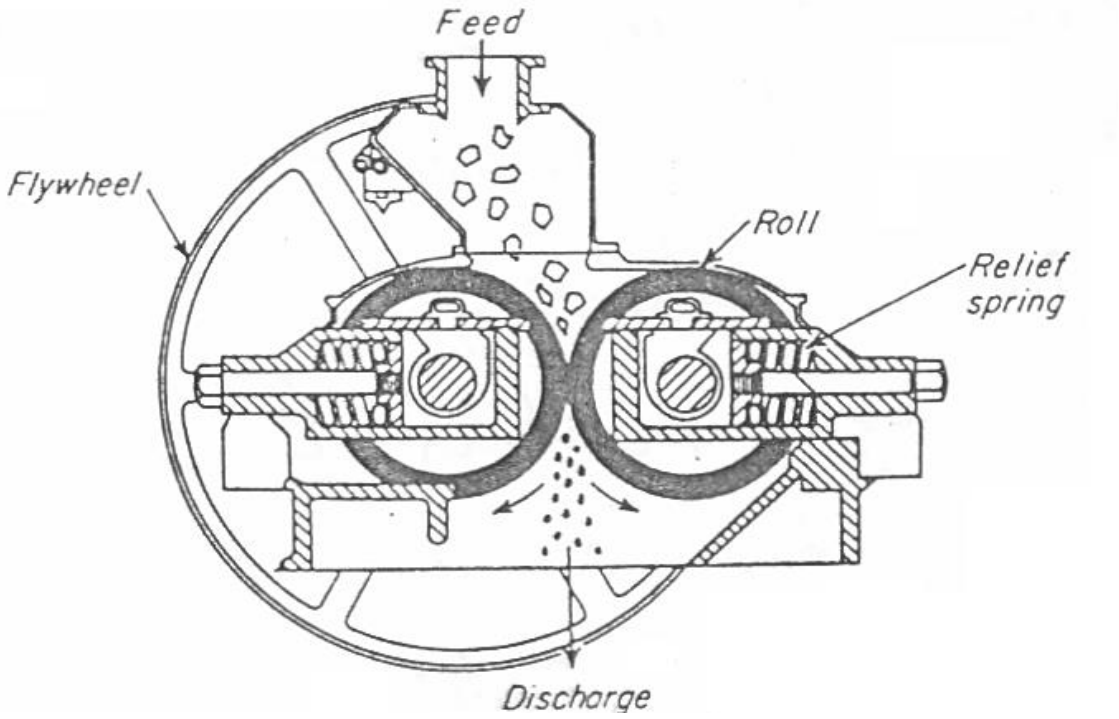


FIGURE 29.3
Gyratory crusher.

The discharge from a Gyratory crusher is continuous instead of intermittent as in a Jaw crusher. The load on the motor is nearly uniform, less maintenance is required than the jaw crusher, and the power required per ton of material crushed is smaller. The capacity of the gyratory crusher varies with the jaw setting, the impact strength of the feed, and the speed of Gyration of the machine,. The capacity is almost independent of the compressive strength of the material being crushed. This has a large capacity unit area of grinding surface,. This will not take such large feed as a jaw crusher with the same through put but it will give a rather finer and more uniform product. Because the capital cost is high, the crusher is suitable only where large quantities of material are to be handled.

Smooth Roll Crushers:

Two heavy smooth faced metal rolls turning m parallel horizontal axis are the working elements of smooth roll crusher.



Particles of feed caught between the rolls are broken in compression and drop out below. The rolls turn toward each other at the same speed. They have relatively narrow faces and are large in diameter so that they can 'nip' moderately large lumps. Roll speeds range from 50 – 300 rpm. Smooth roll crushers are secondary crushers with feeds 12 to 75 mm (1/2 to 3 in) in size and products 12 mm(1/2 in) to about 1 mm. (The limiting size D_p max of the particle that can be ripped by the rolls depends on the co-efficient of friction between the particle and the roll surface, but in most cases it can be estimated by using,

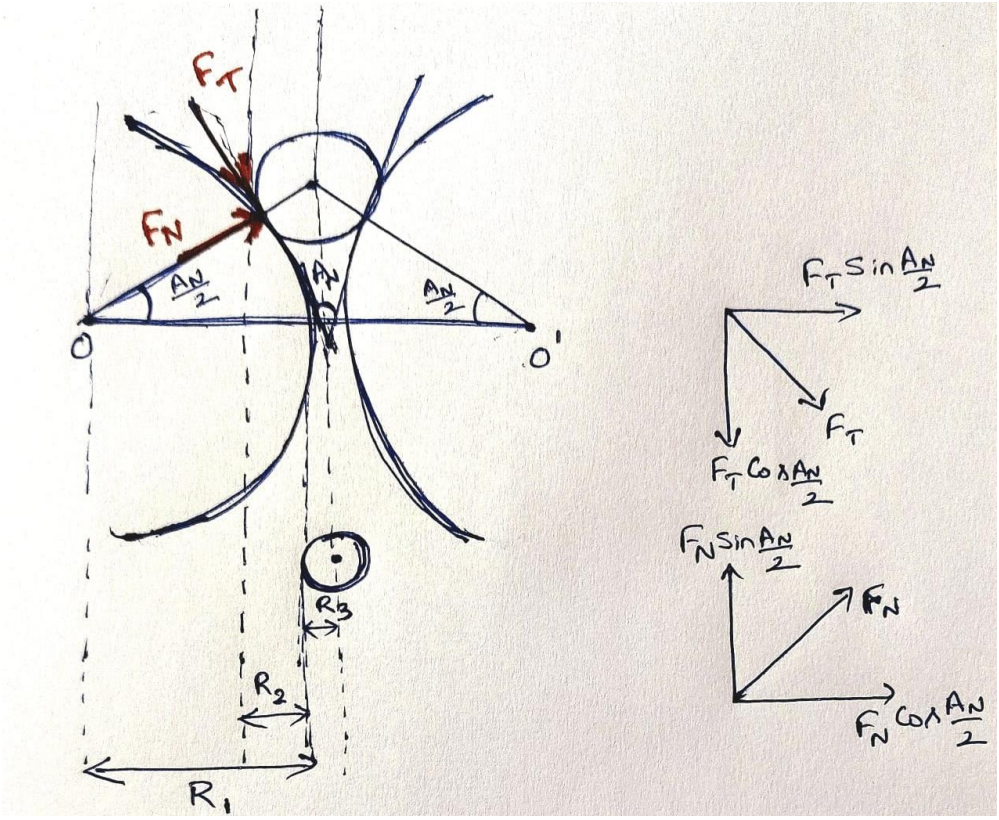
$D_{p\,max} = 0.04\ R + d.$

R – Rollradius, d = ½ the width of the gap between the rolls.

The maximum size of the product is approximately equal to 2d.

The particle size of the product depends on the spacing between the rolls, as does the capacity of a given machines. Smooth roll crushers give few fines of virtually no oversize. They opera more effectively when set to give a reduction ratio of 3 or 4 to 1 i.e., the maximum particle diameter of the product is one-third or one fourth that of the feed. To allow unbreakable material to pass through without damaging the machine, at least one-roll must be spring mounted. Roll crushers are suitable for effecting only it a small size reduction in a single operator and it is therefore common to employ a number of pairs of rolls, one alone the other.

Angle of NIP of Roll Crusher:



$R_1 : Radius\ of\ Roller$
$R_2 : Radius\ of\ Feed$
$R_3 : Radius\ of\ Product$
$A_N : Angle\ of\ Nip$

The angle of nip A_N is the angle between the roll faces at the level where they will just take hold of a particle and draw it into the crushing zone.

In figure let O and O' be the centers of the two rolls of a pair, and There is a certain force F_N acting on the particle and this force makes an angle $\frac{A_N}{2}$ with the line OO' and tends to expel the particle from the rolls This force F_N may be resolved into two components as shown in Figure.

There is also a force F_T tending to draw the particles between the rolls. This force F_T may be resolved into two forces as shown in figure.

The force $F_T Cos\frac{A_N}{2}$ and $F_N Sin\frac{A_N}{2}$ are the two opposing forces. For the particle to be crushed $F_T Cos\frac{A_N}{2}$ should be greater than $F_N Sin\frac{A_N}{2}$.

$$F_T Cos\frac{A_N}{2} > F_N Sin\frac{A_N}{2}$$

$$\frac{F_T}{F_N} > \frac{Sin\frac{A_N}{2}}{Cos\frac{A_N}{2}}$$

$$\frac{F_T}{F_N} = \frac{(Tangential\ Force)}{(Normal\ Force)} = \mu = Coefficient\ of\ Friction$$

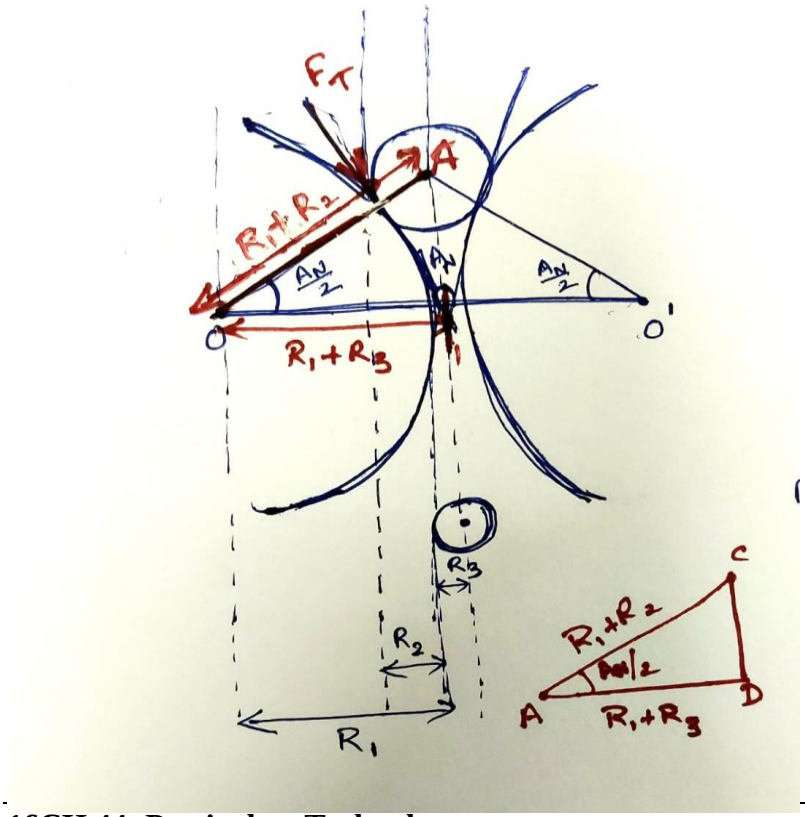
Coefficient of friction (μ) is a dimensionless number that is defined as the ratio between friction force and normal force

$$\mu > tan\frac{A_N}{2}$$

In other words the $tan\frac{A_N}{2}$ must be less than the coefficient of friction. When μ = tan α the particle is just caught between the rolls and ready to be drawn into the crushing zone, and the angle made is angle of nip A_N. Half the angle of NIP is $\frac{A_N}{2}$. The coefficient of friction varies with different materials, but it has been found that an average value of the angle $\frac{A_N}{2}$ taken from practice is about 16°..

Relation between angle of NIP, Dia. of Rolls, radius of Feed and Product:

There is a definite relation between the diameters of rolls, feed and product . In the above figure let R_1 the radius of the roll, R_2 be radius of the feed particle, and R_3 the radius of the largest possible particle in the product (half the distance between the rolls). Then in the triangle OAO' , angle O'OA is $\frac{A_N}{2}$



$$Cos\frac{A_N}{2} = \frac{R_1 + R_3}{R_1 + R_2}$$

For average conditions $\frac{A_N}{2}$ is 16° and $Cos\ 16$ is 0.961 hence

$$0.961 = \frac{R_1 + R_3}{R_1 + R_2}$$

Capacity of Roll Crushers:

It is assumed that the materials to be crushed moves like a continuous ribbon whose width is the width of the roll and whose thickness is the clear opening between the rolls.
$\therefore$ the theoretical volume of material crushed.

$$\textbf{Capacity of Rolls} = \pi DN\ W\ 2d \quad m^3/hr$$

Where

$D: Dia\ of\ roll, m$

$N: Number\ of\ \frac{revolutions}{hr}$

$W: Width\ of\ rolls, m$

$2d: Gap\ between\ the\ rolls$, m

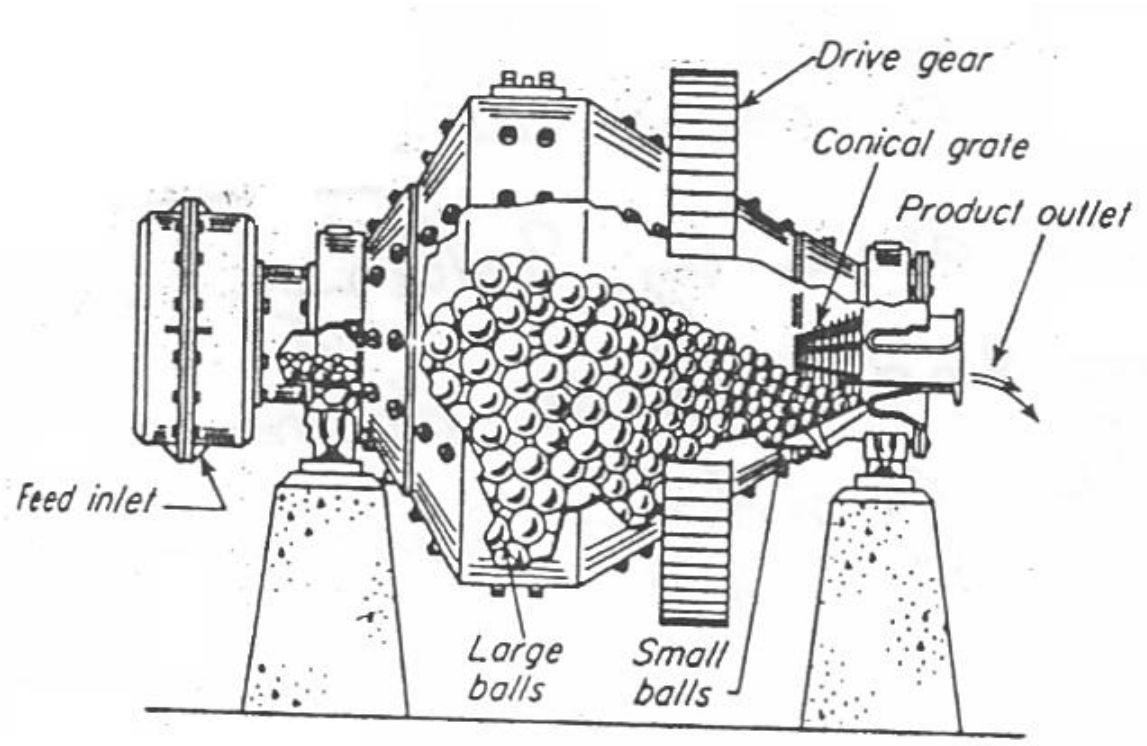
Mass of material crushed = volume x density

Note: The actual capacity is 1/3 rd to 1/10 the of this theoretical capacity.

Ball Mill:

Ball Mill (Tumbling Mill) Revolving Mill:

A cylindrical shell slowly turning about a horizontal axis and filled to about half its volume with a solid grinding medium forms a tumbling mill. The shell is usually steel, lined with high-carbon steel plate, porcelain, silica rock or rubber. The grinding medium is metal rods is a rod mill, lengths of chain or balls of metal, rubber or wood in a ball mill, flint pebbles or porcelain or zircon spheres in a peddle mill. For intermediate and fine reductions of abrasive materials tumbling mills are un equaled.



Tumbling mills may be continuous or Batch. In a Batch machine a measured quantity of the solid to be ground is loaded into the mill through an opening in the shell. The opening is then closed and the mill turned on for several hours it is then stopped and the product is discharged. In a continuous mill the solid flows steadily through the revolving shell, entering at one end through a hollow Trunnion and leaving at the other end through the trunnion or through peripheral openings in the shell.

In all tumbling mills, the grinding elements are carried up the side of the shell nearly to the top from whence they fall on the particles underneath. The energy expended in lifting the grinding units is utilized in reducing the size of the particles. In some tumbling mills as in a rod mill. Much of the reduction is done by rolling compression and by attrition as the rods slide downward and roll over one another. The grinding rods are usually steel, 25 – 125 mm id dia. With several sizes present at all times in any given mill. The rods extend the full length of the mill. Rod mills are intermediate grinders, reducing a 20 mm feed to perhaps 10 – mesh often preparing the product from a crusher for final reduction in a Ball mill. They yield a product with little oversize and a minimum of fines.

In a Ball Mill or pebble mill most of the reduction is done by impact as the balls or pebbles drop from near the top of the shell. In a large ball mill the shell might be 3 m in dia. and 4.25 m long. The balls are 25- 125 mm in dia. The pebbles in a pebble mill are 50 – 175 mm in size. A tube mill is a continuous mill with a long cylindrical shell, in which material is ground for 2 to 5 times as long as in the shorter ball mill. Tube mills are excellent for grinding to very fine powders in a single pass where the amount of energy consumed is not of primary importance. Putting slotted transverse partition in a tube mill connects it into a compartment mill. One compartment may contain large balls, another small balls, and a third pebbles. Thus segregation of the grinding media into elements of different size and weight aids considerably in avoiding wasted work,

for the large heavy balls break only the large particles, without interference by the fines. The small, light balls fall only on small particles, not on large lamps they cannot break.

Action in Revolving Mills (Tumbling Mills):

The load of balls in a ball or tube mill is normally such that when the mill is stopped. The balls occupy one half, the volume of the mill. The void fraction in the mass if balls, when at rest is typically 0.40. The grinding may be done with dry solids, but more commonly the bed is a suspension of the particles in water. This increases both the capacity and efficiency of the mill. Discharge openings at appropriate positions control the liquid bend in the mill, which should be such that the suspension just fills the void space in the mass of balls.

When the mill is rotated, the balls are picked up by the mill wall and carried nearly to the top, where they break contact with the wall and fall to the bottom to be picked up again. Centrifugal force keeps the balls in contact with the wall, and also with each other during the upward movement. While in contact with the wall, the balls do some grinding by slipping and rolling over each other, but most of the grinding occurs at the zone of impact, where the free-falling balls strike the bottom of the mill.

The faster the mill is rotated, the farther the balls are carried up inside the mill and greater the power consumption. The added power is profitably used because the higher the balls are when they are released, the greater the impact at the bottom and the larger the productive capacity of the mill if the speed is too high, however, the balls are carried over and the mill is said to be centrifuging. The speed at which the centrifuging occurs is called **Critical Speed**. Little or no grinding is done when a mill is centrifuging and operating speeds must be less than the critical.

The speed at which the outermost balls loose contact with the wall of the mill depends on the balance between gravitational and centrifugal forces. Thus can be shown a follows.

Forces on ball in ball mill:

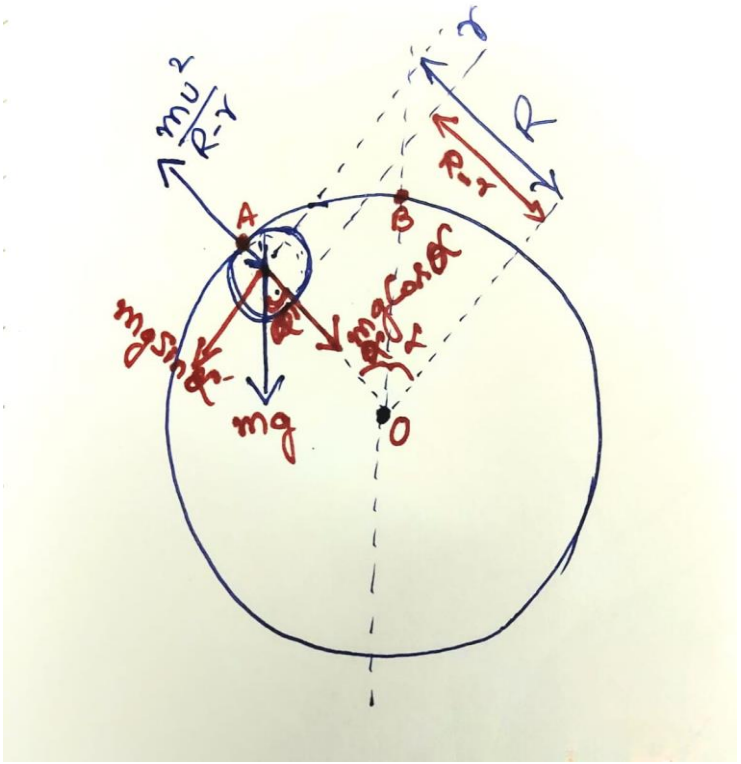
Consider a ball at point A on the periphery of the mill.

$R: Radius\ of\ the\ mill$
$r: Radius\ of\ ball$
$R - r : Distance\ between\ the\ centres\ of\ mill\ and\ Ball.$
Let AO make an angle α with the vertical
$m: Mass\ of\ the\ ball$
$n: Rotational\ speed$
$U: Peripheral\ speed\ of\ the\ center\ of\ the\ ball, U = 2\pi n\ (R - r)$
$Force\ due\ to\ gravity\ on\ the\ ball = mg$

Two forces act on the ball namely:

1. $Centrifugal\ Force: \dfrac{mU^2}{(R - r)}$
2. $Centripetal\ Component: mgCos\alpha$

The centripetal component of the force of gravity is mg cosα and this force opposes the centrifugal force. As long as the centrifugal force exceeds the centripetal force, the particle will not break contact with the wall. As the angle α decreases, however the centripetal force increases and unless the speed exceeds the critical a point is reached where the opposing forces are equal and the particle is ready to fall away. The angle at which this occurs is found by equating the two forces giving



$$\frac{mU^2}{(R-r)} = mgCos\alpha \text{ - - - - - - - - - - - - - - - - - (1)}$$

We Know $U = 2\pi\ (R-r)n$, Substitute for U in (1)

$$\frac{(2\pi n\ (R-r))^2}{(R-r)} = gCos\alpha$$

$$4\pi^2 n^2\ (R-r) = gCos\alpha$$

$$n^2 = \frac{gCos\alpha}{4\pi^2(R-r)}$$

$$n = \frac{1}{2\pi}\sqrt{\frac{gCos\alpha}{(R-r)}}$$

At critical speeds, α = 0, $Cos\alpha = 1$ and n becomes the critical speed N_C, then

$$N_C = \frac{1}{2\pi}\sqrt{\frac{g}{(R-r)}}$$

Tumbling mills run at 65 to 80 percent of the critical speed, with the lower values for wet grinding in viscous suspensions.

Cascading and Cataracting :

Cascading and Cataracting are the terms applied to the motion of the grinding media cascading applies to the rolling of balls or pebbles from the top to the bottom of the heap. Cataracting refers to the throwing of the balls through the air to the toe of the heap.

Advantages of Ball Mill:

1. The mill can be used wet or dry.

2. The cost of installation and of power are low.

3. The ball mill can be used with an inert atmosphere and therefore can be used for the grinding of certain explosive material.

4. The grinding medium is cheap.

5. The mill is suitable for materials of allow degree of hardness.

6. It can be used for batch or continuous operator.

7. It can be used for closed or open circuit grounding.

Advantages of Wet Grinding:

1. The power consumption is less by about 20-30%.

2. The capacity of the plant is increased.

3. Removal of the product is facilitated and the amount of fines reduced.

4. Dust formation is eliminated.

5. The solids are more easily handled.

The disadvantages are that the grinding medium will generally be about 20% greater and it may be necessary to dry the product

Size range of feed and product of size Reduction Equipments:

Equipment	Feed Size	Product size
Coarse Crusher	60'' – 1 1/2 ''	2'' – ¼ ''
Intermediate Crusher	2'' – ¼ ''	¼ '' – 200 mesh
Fine Crusher	¼ '' – 1/8 ''	About 200 mesh
Ultra Fine	80 mesh	Down to 10^{-6}m.

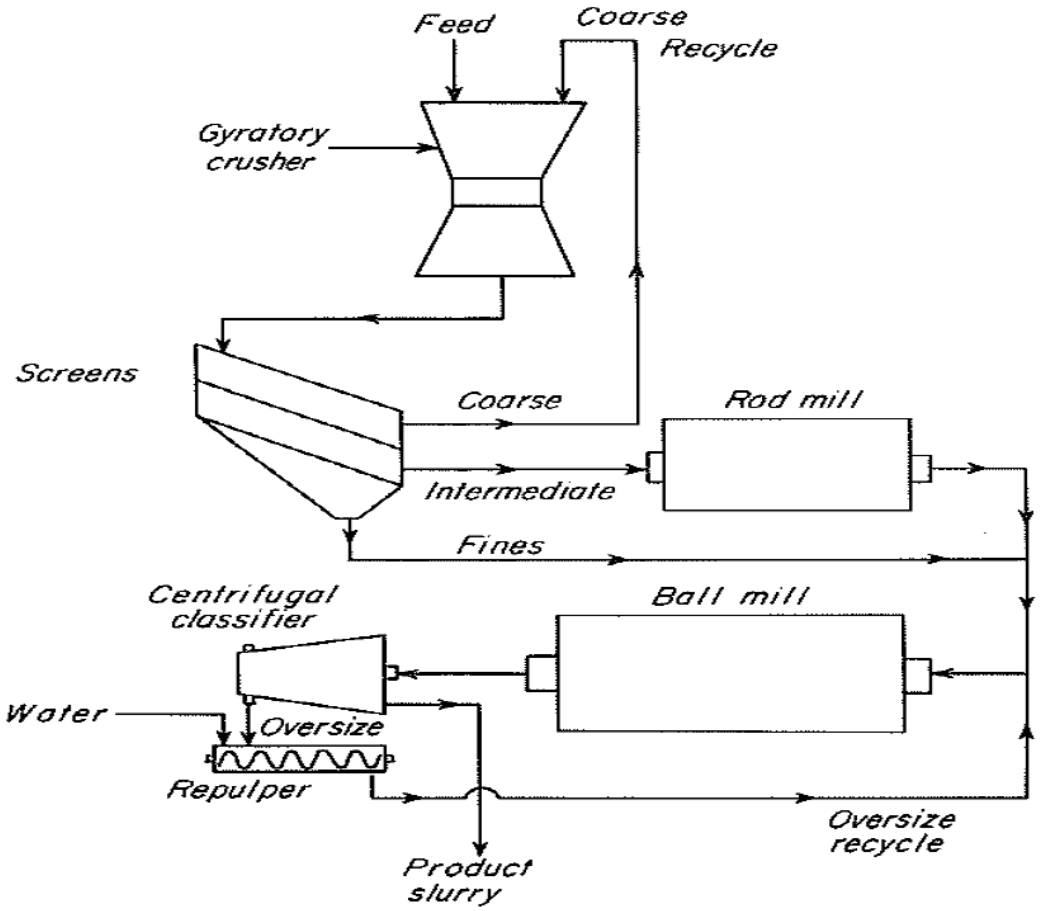
A greater size reduction ratio can be obtained in fine crushers than in coarse crushers.

Methods of Operating Crushers:

The crushers may be operated under free crushing or choke feeding and also either open circuit or closed circuit grinding . If the crusher is fed at a low rate the material crushed will leave the crusher immediately this is called Sew crushing. If the crusher is fed at high rate of solid the material crushed cannot move freely through the crusher and stay in the crusher for longer time and get crushed again. This type of feeding is called Choke Feeding. During choke feeding energy is last because of cushioning effect.

Open circuit and Closed circuit Operation:

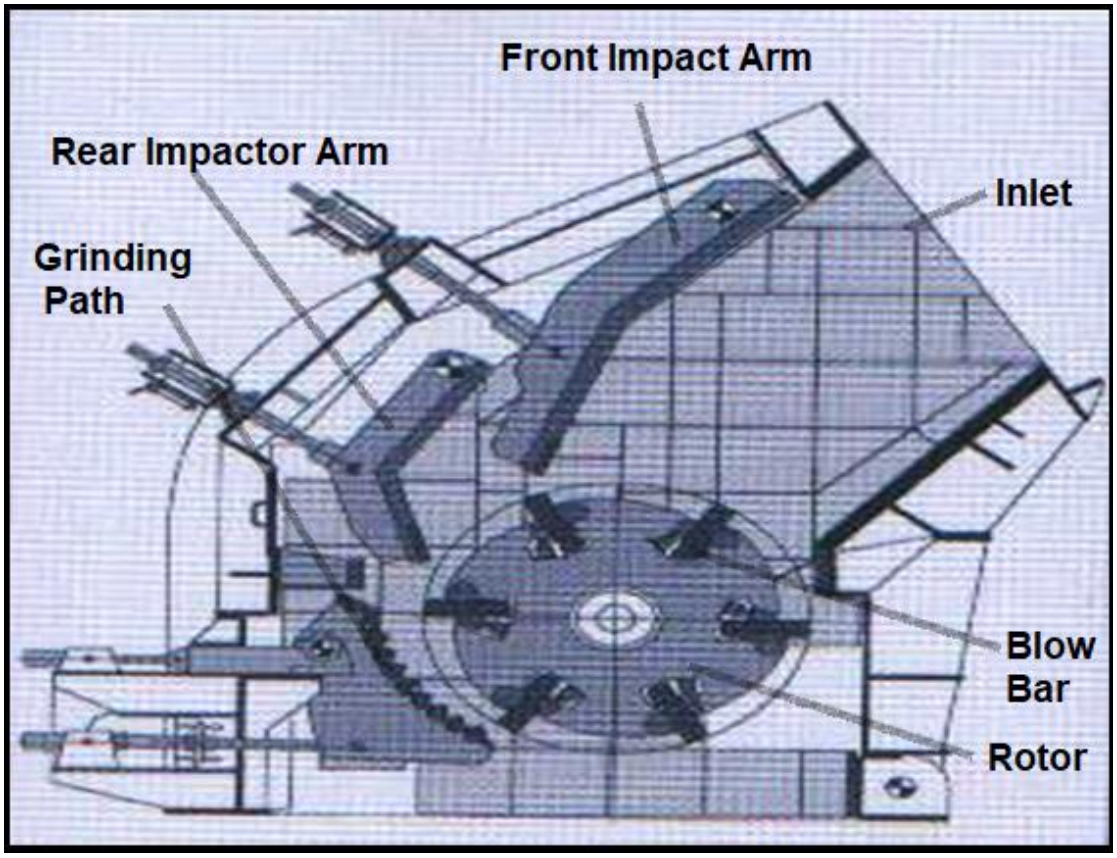
In many mills the feed is broken into particles of satisfactory size by passing it once through the mill. When no attempt is made to return the oversize particles back to the machine for further reduction the mill is said to be operating in open circuit. This may require excessive amounts of pore for much energy is wasted in re-grinding particles that are already fine enough. Thus it is often economical to remove partially ground material from the mill and pass it through a size separation device. The undersize becomes the product and the oversize is returned to be reground. The separation device is sometimes inside the mill, as in ultrafine grinders or as is more common, it is outside the mill. Closed circuit operation is the term applied to the action of a mill and seperator connected so that the oversize particles are returned to the mill.



The above fig. Shows a typical set of size reduction machines and separators operating in closed circuit. [For coarse particles the separation device is a screen or Grizzly for fine powders it is some form of classifier.

The product from the Gyratory crusher is screened into 3 fractions, fines intermediate and oversize. The oversize is sent back to the Gyratory, the fines are fed directly to the final reduction unit, a ball mill. Intermediate particles are broken in a rod mill before they enter the ball mill. In the arrangement shown in the diagram the ball mill is grinding wet. i.e. water is pumped through the mill with the solid to carry the broken particles to a centrifugal classifier The classifier throws down the oversize into a sludge throws down the oversize into a sludge, which is repulped with more water returned to the mill. The undersize, or product, emerges from the classifier as a slurry containing particles of acceptable size. Although screens are simpler to operate them classifiers, they cannot economically make separations when the particles are smaller than about 150-200 mesh. It is the over grinding of precisely these fine particles that results in excessive consumption of energy. Closed circuit operation is therefore of most value in reduction to fine and ultra fine sizes, which demands that the separation we done by wet classifier or air separators. Energy must of course be supplied to drive the conveyors and separators in a closed-circuit system, but despite this, the reduction in total energy requirement over open circuit grinding often reaches 25%.

Impact Crusher:
An impact crusher is a machine that uses striking as opposed to pressure to reduce the size of a material. Impact crushers are designated as a primary, secondary, tertiary or quarternary rotor crusher depending on which processing stage the equipment is being utilized.



Types Of Impact Crushers:

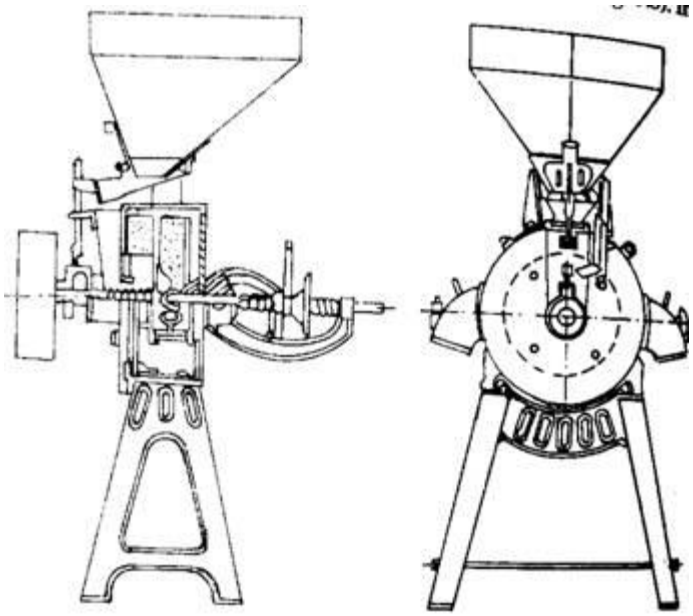
- Horizontal shaft impactors (HSI): Mega Slam Primary Impact Crusher & Grand Slam Secondary Impact Crusher
- Cage mill pulverizers
- Vertical shaft impactors (VSI)

Feed enters the crushing chamber and meets the breaker bars or plates propelling feed against the breaker plates resulting in impact reduction. There are no screens or grates holding material inside impact crushers, so material is efficiently processed at high rates for low costs.

Benefits of Impact Crushers:
Impact crushers reduce mineral materials such as concrete, asphalt and natural rock in size to produce a valuable commodity product. A fast spinning rotor throws the material against a solid stationary impact wall. The striking and impacting causes the material to shatter into smaller pieces. The result is a very homogenous and cubical product leaving the crusher box.
Ideal for producing a spec product without the need of a primary jaw crusher.
Material produced by an impact crusher is easier to grade and compact making it perfect for base and subbase.

Attrition mill

- Also known as burr mill
- Grains are rubbed between the grooved flat faces of rotating circular disks
- Axis of the roughened disks may be horizontal or vertical
- One plate is stationary and fixed with the body of the mill while the other one is rotating disk
- Material is fed between the plates and is reduced by crushing and shear
- Mills with different patterns of grooves, corrugations on the plates perform a variety of operations
- Overfeeding
    - lowers grinders performance
    - Increases heat generation during milling
- Disks are 20-137 cm in dia and operated at 350 to 700 rpm



- Used for making whole grain and dehusked grain flour
- Use in spice grinding is limited
- Double runner disks type attrition mills are also available
- Used for grinding of soft materials
- Both disks are driven at high speed in opposite direction
- Operated between 1200 to 7000 rpm
- Capacity is large

Salient features

- Fineness of grinding is controlled by the type of plates and the gap between them
- Spacing between the plates is adjustable
- Arrangement is spring loaded
    - to avoid damage to plates in case of overloading
    - to overcome the damage to plates by foreign material coming along with the feed
- Lower initial cost
- Lower power requirements

PROBLEMS:

1. What should be the diameter of a set of rolls to take the feed size equivalent to 4 cm and crush it to 12.5 mm sphere if the coefficient of friction is 0.29.

Solution:

$R_2 : Radius\ of\ Feed = \frac{4}{2} = 2\ cm = 2\ x\ 10^{-2} m$

$R_3 : Radius\ of\ Product\ = \frac{12.5}{2} = 6.25\ mm = 6.25\ x\ 10^{-3} m$

$R_1 : Radius\ of\ Roller\ =?$

$\mu = 0.29$

$$Cos\frac{A_N}{2} = \frac{R_1 + R_3}{R_1 + R_2}$$

$$\mu = tan\frac{A_N}{2}$$

$$tan^{-1}0.29 = \frac{A_N}{2}$$

$$\frac{A_N}{2} = 16.17$$

$$Cos16.17 = \frac{R_1 + R_3}{R_1 + R_2} = \frac{R_1 + 6.25\ x\ 10^{-3}}{R_1 + 2\ x\ 10^{-2}}$$

$$Cos16.17 = \frac{R_1 + 6.25\ x\ 10^{-3}}{R_1 + 2\ x\ 10^{-2}}$$

$$R_1 = 32.76\ cm\ or\ 0.3276\ m$$

2. The diameter of set of rolls is 1.4 m and takes a feed of size equivalent to 5 cm. If the Angle of Nip is 30º, What is the maximum size of Product.
Solution:

$R_1 : Radius\ of\ Roller = \frac{1.4}{2} = 0.7\ m$

$R_2 : Radius\ of\ Feed = \frac{5}{2} = 2.5\ cm = 2.5\ x\ 10^{-2}\ m$

$R_3 : Radius\ of\ Product = ?$

$A_N : Angle\ of\ Nip\ =\ 30^o\ and\ \frac{A_n}{2} = 15^o$

$$Cos15 = \frac{R_1 + R_3}{R_1 + R_2} = \frac{0.7 + R_3}{0.7 + 2.5\ x\ 10^{-2}}$$

$$Cos15 = \frac{0.7 + R_3}{0.7 + 2.5\ x\ 10^{-2}}$$

$$R_3 = 0.029\ m$$

Size of Product $0.029\ x\ 2 = 0.052\ m$

3. A pair of crushing rolls 1m dia are so set that a gap of 12.5 mm is between them. What is the size of largest particles that can be fed to rolls. Angle of Nip is 31º. If the rolls are 0.4 m wide run at 100 rpm, Calculate the through put per hour in tonnes of feed. Density 2330 kg/m^3. Efficiency is 12%.

Solution:

$$Cos\frac{A_N}{2} = \frac{R_1 + R_3}{R_1 + R_2}$$

$R_1 : Radius\ of\ Roller = \frac{1.0}{2} = 0.5\ m$

$R_2 : Radius\ of\ Feed = \frac{5}{2} = 2.5\ cm = 2.5\ x\ 10^{-2}\ m$

$R_3 : Radius\ of\ Product = ?$

$A_N : Angle\ of\ Nip\ =\ 31^o\ and\ \frac{A_n}{2} = 15.5^o$

$\eta = 12\% = 0.2$
$W = 0.4$
$D = 1m$
$2d = 12.5\ x\ 10^{-3}$m
$N = 100\ rpm = 100x\ 60\ rps$
$\rho = 2330\ kg/m^3$

$$Cos\frac{A_N}{2} = Cos15.5^o = \frac{0.5 + R_3}{0.5 + 2.5\ x\ 10^{-3}}$$

$$R_3 = 0.0254\ m$$

Size of the largest particle = 2 x 0.0254 = 0.0508m

Capacity of Rolls = πDN W 2d ρ η m^3/hr

$=\pi\ x1.0\ x100x60\ x0.4\ x12.5\ x\ 10^{-3}x2330\ x\ 0.12$

$$= 26351.68kg/hr$$

$$26.35\ T/hr$$

4. A ball mill 1.2 m dia is being run at 50 rpm. What is the size of the ballswe have to employ for proper grinding.

Solution:

Speed of ball mill: 50 rpm (Assume optimum speed)
Critical speed of ball mill $=\frac{50}{0.7} = 71.43\ rpm$

$$N_C = Critical\ speed = \frac{71.43}{60} = 1.1405\ rps$$

$$Radius\ of\ ball\ mill\ (R) = \frac{1.2}{2} = 0.6\ m$$

Radius of ball $r = ?$

$$N_C = \frac{1}{2\pi}\sqrt{\frac{g}{(R-r)}}$$

$$1.1405 = \frac{1}{2\pi}\sqrt{\frac{9.81}{(0.6-r)}}$$

$$r = 0.425\ m$$

5. What is the critical speed in rpm of a ball mill 1000 mm ID with 75 mm diameter.

Solution:

$N_C = Critical\ speed = ?$

$$Radius\ of\ ball\ mill\ (R) = \frac{1000}{2} = 500\ mm$$

Radius of ball $r = \frac{75}{2} = 37.5\ mm = 37.5\ x10^{-3}$

$$N_C = \frac{1}{2\pi}\sqrt{\frac{g}{(R-r)}}$$

$$N_C = \frac{1}{2\ x\ 3.142}\sqrt{\frac{9.81}{(500x10^{-3} - 37.5x10^{-3})}} = 0.733\ rps$$

$$= 0.733\ x\ 60 = 43.98\ rpm$$

6. A ball mill of 120 cm is opened at 50 rpm. It is found that it is not working satisfactory. Explain the reason for this and suggest operating condition. The dia of balls use are 70 mm.

Solution:

$$N_C = \frac{1}{2\pi}\sqrt{\frac{g}{(R-r)}}$$

$$N_C = \frac{1}{2\ x\ 3.142}\sqrt{\frac{9.81}{(60x10^{-2} - 35x10^{-3})}} = 0.663\ rps$$

$$= 0.633\ x\ 60 = 39.78\ rpm$$

Since the operating speed is more than the critical speed it is not working satisfactorily.

Suggestion:

1. Reduce the speed of the mill below the critical speed
2. Increase the critical speed of the mill by increasing the size of the grinding medium such that N_C is above 50, the present operating speed